

Fig. 2: An actual-size, single-side PCB of the demo circuit for over-voltage protection

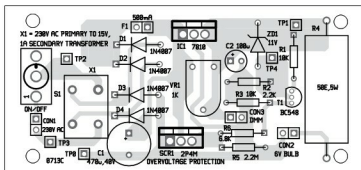


Fig. 3: Component layout for the PCB

Once SCR1 triggers, fuse blows and disconnects the power supply from the device.

For demo, set VR1 at the maximum limit (say, 1k) and switch on the circuit. Using a digital multimeter, measure the output at CON3. It should be around 10.3 V. Now reduce VR1 resistance in steps. At around 800 Ω , the multimeter reads 10.9 V. Reduce the resistance further until SCR1 fires. Experimentally, it was found that at around 680 Ω , the SCR turns on after receiving a triggering pulse and

a heavy current passes through the fuse wire. Due to this, the fuse wire blows and the load disconnects from the supply.

Construction and testing

An actual-size, single-side PCB of the demo circuit for over-voltage protection is shown in Fig. 2 and its component layout in Fig. 3. After assembling the circuit on PCB, enclose it in a suitable box.

To test the circuit for proper functioning, switch on S1 and measure

the input voltage (230V AC) between TP2 and TP3. Also verify the output of IC1 as 10 V at TP1 with respect to TP0. Check voltage variation at the base of transistor T1 corresponding to change in the resistance value of VR1. ●

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